IsDB-BISEW

IT SCHOLARSHIP PROGRAMME

PROJECT PROPOSAL

Online Bus Ticket Management System

A Web-Based Application using PHP and MySQL

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1. Introduction

Buying a bus ticket in many places still means standing in a long queue at a counter or calling an agent. This takes time and often causes problems such as sold-out seats, wrong bookings, and no record of past trips. As more people use the internet and mobile phones, an online ticket system is now a basic need for bus travel.

This proposal describes the Online Bus Ticket Management System (OBTMS). It is a web-based application where passengers can search for buses, check seat availability, book tickets, and manage their bookings from any device. Bus companies and administrators can use the same system to manage buses, routes, schedules, and sales reports. The system will be built with PHP and MySQL.

2. Problem Statement

Most small and medium bus services still use manual or partly manual methods to sell tickets. The main problems are:

- Passengers must visit a counter to book, which costs time and travel effort.
- There is no live view of seat availability, so double booking can happen.
- Paper records are easy to lose, hard to search, and can contain errors.
- Cancelling or changing a ticket is slow and often unclear.
- Owners find it hard to create sales and passenger reports.

A simple, secure, and low-cost online system is needed to solve these problems.

3. Objectives of the Project

3.1 Main Objective

To design and develop a web-based Online Bus Ticket Management System that makes ticket booking fast, easy, and reliable for both passengers and bus operators.

3.2 Specific Objectives

1. To let passengers register, log in, and search buses by route, date, and time.
2. To show live seat availability using a seat layout.
3. To allow online ticket booking, payment recording, and ticket cancellation.
4. To let the admin manage buses, routes, schedules, fares, and users.
5. To generate printable tickets and useful reports.
6. To keep user data safe using secure login and input validation.

4. Scope of the Project

4.1 In Scope

- User registration, login, and profile management.
- Bus search, seat selection, and ticket booking.
- Booking history, ticket cancellation, and ticket printing (PDF or print view).
- Admin panel for bus, route, schedule, fare, booking, and user management.
- Sales and booking reports.
- Responsive design that works on desktop, tablet, and mobile browsers.

4.2 Out of Scope

- Real payment gateway integration (a simulated payment record will be used in this version).
- Native mobile applications for Android or iOS.
- GPS-based live bus tracking.

5. Review of Existing System

In the existing manual system, tickets are sold at counters and records are kept in registers or spreadsheets. Some operators take bookings by phone. The table below compares the existing system with the proposed system.

Feature	Existing System	Proposed System
Booking method	Counter or phone call	Online, 24/7
Seat availability	Checked manually	Live seat map
Record keeping	Paper or spreadsheet	Central MySQL database
Cancellation	Visit counter, slow	Online, quick
Reports	Manual, error-prone	Automatic reports
Accuracy	Risk of double booking	Seat locked after booking

Table 1: Existing system vs. proposed system

6. Proposed System

The proposed system is a web application for three types of users. Each user type has its own dashboard and permissions.

- **Passenger:** searches buses, selects seats, books and cancels tickets, and views booking history.
- **Admin:** has full control. Manages buses, routes, schedules, fares, users, bookings, and reports.
- **Counter Staff (optional):** books tickets for walk-in passengers using the same system.

All data is stored in one MySQL database. When a seat is booked, it is marked as unavailable at once, so the same seat cannot be sold twice.

7. System Modules and Features

7.1 Passenger Module

- Sign up, log in, log out, and reset password.
- Search buses by origin, destination, and travel date.
- View bus details: type (AC or Non-AC), departure time, fare, and available seats.
- Select seats from a visual seat layout.
- Confirm booking and receive a ticket with a unique ticket number.
- View booking history and cancel upcoming tickets.

7.2 Admin Module

- Dashboard with total bookings, income, and active buses.
- Add, edit, and delete buses, routes, and schedules.
- Set and update ticket fares.
- View and manage all bookings and users.
- Generate daily, weekly, and monthly reports.

7.3 Ticket and Payment Module

- Generate a unique ticket with passenger name, route, seat number, date, and time.
- Record payment method and payment status (Paid / Pending / Refunded).
- Apply a simple cancellation rule, for example refund before a set time limit.

7.4 Security Module

- Password hashing using PHP password_hash().
- Prepared statements (MySQLi) to prevent SQL injection.
- Input validation and output escaping to prevent XSS.
- Session-based login and role-based access control.

8. System Requirements

8.1 Functional Requirements

1. The system shall allow users to create an account and log in.
2. The system shall let users search buses by route and date.
3. The system shall show available and booked seats for each trip.
4. The system shall let users book and cancel tickets.

5. The system shall generate a printable ticket.
6. The system shall let the admin manage all system data.
7. The system shall generate booking and sales reports.

8.2 Non-Functional Requirements

- **Usability:** simple and clear interface that new users can learn quickly.
- **Performance:** pages should load within a few seconds on a normal connection.
- **Security:** protected passwords, safe queries, and role-based access.
- **Reliability:** no double booking and regular database backup.
- **Compatibility:** works on modern browsers such as Chrome, Firefox, and Edge.

8.3 Hardware and Software Requirements

Item	Requirement
Processor	Intel Core i3 or higher
RAM	4 GB minimum (8 GB recommended)
Storage	500 MB free space for project and database
Operating System	Windows 10/11, Linux, or macOS
Server	Apache (via XAMPP / WAMP / LAMP)
Database	MySQL 5.7 or higher / MariaDB
Language	PHP 8.0 or higher
Browser	Latest Chrome, Firefox, or Edge

Table 2: Hardware and software requirements

9. Technology Stack

Layer	Technology
Front-end	HTML5, CSS3, JavaScript, Bootstrap (responsive design)
Back-end	PHP (core PHP, or a light framework if needed)
Database	MySQL
Web Server	Apache
Tools	VS Code, XAMPP, phpMyAdmin, Git
Design Tools	draw.io or Lucidchart (diagrams)

Table 3: Technology stack

10. System Architecture

The system follows a simple three-tier architecture. Each layer has its own job, which makes the system easy to build, test, and maintain.

Layer	Role	Components
Presentation Layer	What the user sees and uses	HTML, CSS, JavaScript, Bootstrap
Application Layer	Business logic and processing	PHP scripts, session handling, validation
Data Layer	Stores and retrieves data	MySQL database

Table 4: Three-tier architecture

Flow: The user sends a request from the browser. The PHP script on the Apache server checks the request, talks to the MySQL database, and sends the result back as a web page.

11. Database Design

The database will be normalized to reduce repeated data. The main tables are listed below.

Table	Main Fields	Purpose
users	user_id, name, email, phone, password, role, created_at	Stores all user accounts
buses	bus_id, bus_name, bus_number, bus_type, total_seats	Stores bus information
routes	route_id, source, destination, distance	Stores travel routes
schedules	schedule_id, bus_id, route_id, travel_date, departure_time, fare	Stores trips and fares
seats	seat_id, schedule_id, seat_number, status	Tracks seat availability
bookings	booking_id, user_id, schedule_id, ticket_no, total_fare, status, booked_at	Stores ticket bookings
booking_seats	id, booking_id, seat_id	Links bookings to seats
payments	payment_id, booking_id, amount, method, status, paid_at	Stores payment records

Table 5: Main database tables

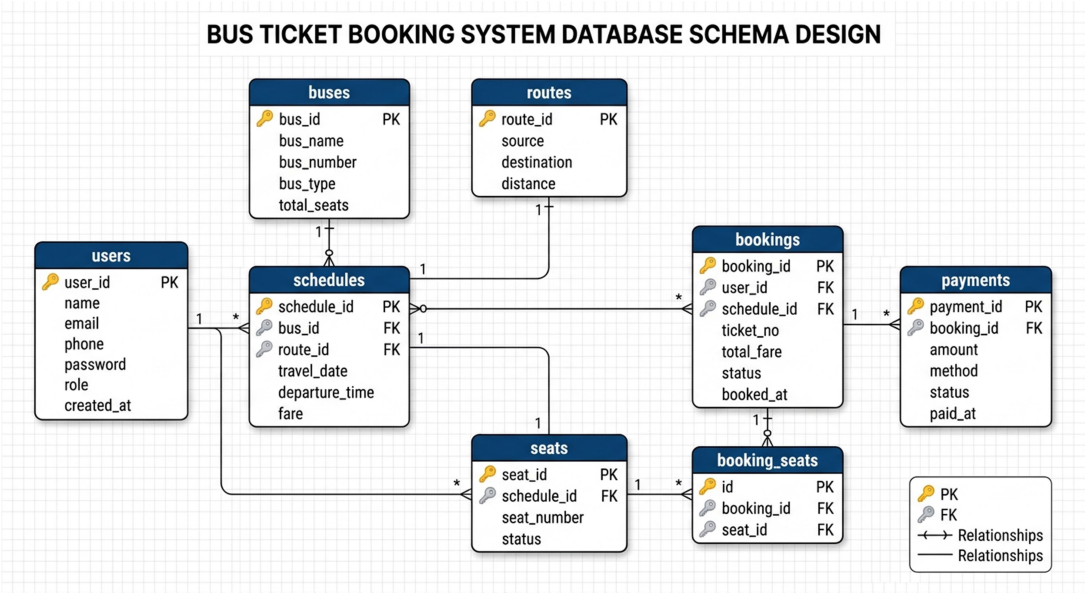


Figure 1: Database schema (entity-relationship diagram)

Key Relationships

- One user can make many bookings (one-to-many).
- One bus can have many schedules, and one route can have many schedules.
- One schedule has many seats.
- One booking can include many seats (through booking_seats).
- One booking has one payment record.

12. Development Methodology

The project will follow the Waterfall model. This model is simple, clear, and suitable for an academic project with fixed requirements. The phases are:

1. Requirement Analysis: collect and study what the system must do.
2. System Design: design the architecture, database, and user interface.
3. Implementation: write code for each module using PHP and MySQL.
4. Testing: test each module and the full system, then fix errors.
5. Deployment: install the system on a local or live server.
6. Maintenance: correct bugs and make small improvements.

Testing Plan

- Unit testing for each function and form.
- Integration testing between modules, such as booking and payment.
- User acceptance testing with sample users.
- Security testing for login, SQL injection, and XSS.

13. Project Timeline

The project is planned for 30 days.

No.	Activity	Duration
1	Requirement analysis and planning	Day 1 – 4
2	System and database design	Day 5 – 8
3	User and admin module development	Day 9 – 15
4	Booking, seat, and payment module development	Day 16 – 22
5	Testing and bug fixing	Day 23 – 26
6	Documentation, deployment, and presentation	Day 27 – 30

Table 6: Project timeline

14. Expected Outcomes

- A working online bus ticket system that is easy to use.
- Less waiting time and fewer booking mistakes for passengers.
- No double booking because of live seat control.
- Better record keeping and quick reports for bus operators.
- Practical experience in PHP, MySQL, and web application development.

15. Limitations and Future Work

15.1 Limitations

- The first version will not connect to a real payment gateway.
- It will be a web application only, not a mobile app.
- Live bus tracking is not included.

15.2 Future Work

- Integrate mobile payment services and card payment gateways.
- Send SMS and email notifications for booking and cancellation.
- Develop Android and iOS apps.
- Add GPS-based bus tracking and QR-code ticket checking.
- Add discount coupons and a rating system for services.

16. Conclusion

The Online Bus Ticket Management System will solve the common problems of manual ticket selling by giving passengers a fast and simple way to book tickets, and giving operators a better way to manage their business. Using PHP and MySQL keeps the project low-cost, easy to develop, and easy to maintain. With careful planning and testing, this project can be completed on time and can serve as a strong base for future improvements.

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